Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

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Summary

Wu and Blei (2025) place a prior over binary selectors $z \in \{0,1\}^p$, where $z_j = 1$ indexes a feature included in the invariant set S^* (Peters et al., 2016). The posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z)/\hat{p}_e(Y_i^e \mid X_{ei}^z)$ rewards selectors for which the pooled conditional $\hat{g}$ dominates each local conditional $\hat{p}_e$, where $\hat{p}_e$ is fit separately per environment e. Consistency (Thm 1) requires $\mu_{\min} > 0$, defined as the minimum KL divergence between $\hat{g}$ and any $\hat{p}_e$ over non-invariant features (Wu and Blei, 2025, Def. 1). Contraction (Thm 2) gives posterior mass on $\{z \neq z^*\}$ scaling as $O(R\, e^{-\kappa n E \mu_{\min}})$, where $R = 2^p$ counts hypotheses, E is the number of environments, and κ is a universal constant. VI-BIP (Wu and Blei, 2025, Sec. 4) approximates the posterior via mean-field variational inference (Blei et al., 2017), optimizing the evidence lower bound (ELBO) using the U2G gradient estimator to handle discrete z.

Critique: One Invariance Condition, Three Algorithms

The most striking observation across this body of work is that Peters et al. (2016), Fan et al. (2024), and Wu and Blei (2025) all enforce the same underlying condition ($P_e(Y \mid X_{S^*})$ is constant across environments e) but operationalize it in structurally incompatible ways that each paper treats as self-contained.

Peters et al. test invariance sequentially, building a confidence set $\hat{S}(\mathcal{E})$ that shrinks monotonically as environments accumulate (Peters et al., 2016, Thm. 1). BIP's Thm 2 is the probabilistic counterpart: rearranging the contraction gives sample complexity $nE \gtrsim O(p/\mu_{\min})$, which is the per-environment requirement for the Peters test to have nontrivial power. BIP thus does not replace ICP; it quantifies ICP's frequentist power curve in a Bayesian language.

NegDRO (Fan et al., 2024) makes a third move. Its minimax problem $\min_b \max_{w \in U(\gamma)} \sum_e w_e \hat{\mathbb{E}}[(Y^e - b^\top X^e)^2]$, where $U(\gamma)$ is the set of signed environment weights with total variation at most γ (Fan et al., 2024, Def. 1), is the distributionally robust framing of Bühlmann (2020) pushed to an adversarial extreme. The critical difference from BIP is structural: BIP treats environments symmetrically inside the product likelihood, while NegDRO can amplify a single harsh environment pair, making it more powerful when one intervention is known *a priori* to be maximally informative. Symmetry is appropriate when intervention quality is unknown and one wants the data to speak; asymmetry (NegDRO) is appropriate when domain knowledge identifies a dominant intervention. This suggests a natural hybrid: use NegDRO to identify the environment pair with the largest empirical $\hat{\mu}_e$, then initialize BIP's prior to concentrate on selectors that perform well under that pair, rather than treating both methods as competing alternatives.

Questions

1. If BIP's $\mu_{\min}$ plays the same role as the power parameter in Peters et al., can one construct a Bayesian power analysis (a prior over $\mu_{\min}$) to decide how many environments to collect before fitting BIP, analogous to classical sample-size planning?
2. VI-BIP's mean-field family $q(z) = \prod_j \mathrm{Bern}(z_j; \pi_j)$ ignores posterior correlation between z_j and z_k. For strongly correlated covariates, does the mean-field gap bias π_j toward 0.5? Would a structured variational family (e.g. an Ising model over z) recover sharper posteriors, and can the U2G estimator extend to that setting?

References

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